

Exercícios Números

2.2 e 2.7

[1] P

[2]

$$P = 660 \text{ W}$$

$$\Delta t = 30 \text{ min} = 0,5 \text{ horas}$$

$$\Delta U_e = P \cdot \Delta t = 660 \cdot 0,5 = 330 \text{ W.h} = 0,33 \text{ kW.h}$$

$$P_{\text{ref}} = 10 \cdot 0,33 = 3,3 \text{ centenas}$$

[3]

$$\bar{I} = \frac{6-3}{2} = 1,5 \text{ A}$$

$$I = \frac{\Delta q}{\Delta t} \Rightarrow \Delta q = 1,5 \times 3 = 4,5 \text{ A.h}$$

$$= 4,5 \cdot 60^2 \text{ A.s} = 16,2 \text{ kC}$$

[4]

$$\Delta q_i = 8 \text{ A.h}$$

$$\bar{I} = 0,05 \text{ A}$$

$$\Delta q_f = 8 - 0,05 \cdot 50 = 5,5 \text{ A.h}$$

$$\Delta t = 50 \text{ h}$$

$$\frac{\Delta q_f}{\Delta q_i} = \frac{5,5}{8} = 0,688 \rightarrow 68,8\%$$

[5]

$$4,0 \times 10^{18} \text{ elétrons}$$

$$3,5 \times 10^{18} \text{ prótons}$$

$$q_{\text{prótons}} + q_{\text{elétrons}} = 5,5 \times 10^{18}$$

$$q_{\text{res}} = 5,5 \times 10^{18} \cdot \frac{e}{1,6 \times 10^{-19} \text{ C}} = 0,88 \text{ C}$$

$$\bar{I} = \frac{\Delta q}{\Delta t} = \frac{0,88}{1} = 0,88 \text{ A}$$

Problemas

1)

$$v_0 = 0 \text{ m/s}$$

$$\Delta V = 220 \text{ V}$$

$$m_e = 9,109 \times 10^{-31} \text{ kg}$$

$$\begin{aligned} \Delta U_e &= |q \cdot \Delta V| = |e \cdot \Delta V| = |1,6 \times 10^{-19} \cdot 220| \\ &= 3,52 \times 10^{-17} \text{ J} \quad (e \cdot V = \text{J}) \end{aligned}$$

$$\Delta U_e = \Delta E_c \Leftrightarrow 3,52 \times 10^{-17} = \frac{1}{2} \cdot 9,109 \times 10^{-31} \cdot v^2 \Leftrightarrow$$

$$\Leftrightarrow v = \pm 8,79 \times 10^6 \text{ m/s}$$

2)

$$v_0 = 0 \text{ m/s}$$

$$e = 1,6 \times 10^{-19} \text{ C}$$

$$\Delta V = 4 \text{ kV}$$

$$m_e = 9,109 \times 10^{-31} \text{ kg}$$

$$a) \quad \Delta U_e = |q \cdot \Delta V| = |e \cdot \Delta V| = 4000 \text{ e.V} = 6,4 \times 10^{-16} \text{ J} \quad (e \cdot V = \text{J})$$

Não existem forças dissipativas, logo $\Delta E_{mec} = 0$;

A energia potencial gravítica é muito pequena comparada com ΔU_e logo pode ser assumida zero.

$$\Delta E_{mec} = 0 \Leftrightarrow \Delta U_e = \Delta E_c \Leftrightarrow 6,4 \times 10^{-16} = \frac{1}{2} \cdot 9,109 \times 10^{-31} \cdot v^2 \Leftrightarrow$$

$$\Leftrightarrow v = 3,75 \times 10^7 \text{ m/s} \quad \checkmark$$

$$b) \quad \bar{E} = \frac{\Delta V}{\Delta r} \left(\frac{V}{m} \right) = \frac{4k}{0,8} = 5000 \text{ V/m} \quad \checkmark$$

$$3) \quad \text{Carga}_{100\%} = 250 \text{ A.h}$$

$$a) \quad 60\% \quad \text{Carga}_{60\%} = \text{Carga}_{100\%} \cdot 0,6 = 150 \text{ A.h} = 150 \cdot 60^2 \text{ A.s} = 5,40 \times 10^5 \text{ C} \quad \checkmark$$

$$b) \quad \bar{I}_{\text{equivalente}} = \frac{3+7}{2} = 5 \text{ A}$$

$$\text{Carga}_{\text{equivalente}} = 5 \cdot 6 = 30 \text{ A.h}$$

$$\text{Carga}_{60\%} + \text{Carga}_{\text{equivalente}} = 150 + 30 = 180 \text{ A.h}$$

$$\frac{\text{Carga}_{\text{equivalente}}}{\text{Carga}_{100\%}} = \frac{180}{250} = 72\% \quad \checkmark$$

$$4) \quad \Delta V = E = 1,2 \text{ V}$$

$$\text{Carga}_{100\%} = q_{\text{max}} = 2,3 \text{ A.h}$$

$$\Delta U_{\text{elec}} = q_{\text{max}} \cdot E = 2,3 \times 1,2 = 2,76 \text{ A.h.V} = 2,76 \times 60^2 \text{ A.s.V} = 9,94 \times 10^3 \text{ C.V} = 9,94 \times 10^3 \text{ J} \quad \checkmark$$

$$5) \quad I = 40 \text{ mA} \quad \text{ou} \quad q_{\text{max}} = 8 \text{ A.h}$$

$$\Delta V = 3 \text{ V} \quad \Delta V = 2 \times 1,5 = 3 \text{ V}$$

$$P = I \cdot \Delta V = 0,04 \cdot 3 = 0,12 \text{ W}$$

$$I = P / \Delta V = 0,12 / 3 = 0,04 \text{ A}$$

$$\Delta t = \frac{q_{\text{max}}}{I} = \frac{8}{0,04} = 200 \text{ h} \quad \checkmark$$

6

$$I_{\min} = 30 \text{ A}$$

$$\Delta V = 230 \text{ V}$$

$$P_{\min} = \Delta V \cdot I_{\min} = 230 \times 30 = 6,9 \text{ kW}$$

7

$$I = 20 + 3 \cdot t^2 \text{ mA} \quad (t \text{ in s})$$

$$\begin{aligned} \text{a)} \quad | \Delta q | &= \int_{t_0}^{t_1} I \cdot dt = \int_0^{10} 20 + 3 \cdot t^2 dt = \left[20 \cdot t + t^3 \right]_0^{10} = 200 + 1000 \\ &= 1,2 \text{ A} \cdot \text{s} = 1,2 \text{ C} \end{aligned}$$

$$\text{b)} \quad \bar{I} = \frac{| \Delta q |}{\Delta t} = \frac{1,2}{10} = 0,12 \text{ A}$$

8

$$\Delta V = 1,5 \text{ V}$$

$$9,6 \times 10^{27} \text{ electrons}$$

$$\Delta t = 2 \text{ h}$$

$$\text{a)} \quad \bar{I} = \frac{| \Delta q |}{\Delta t} = \frac{| 9,6 \times 10^{27} \cdot 1,6 \times 10^{-19} |}{(2 \cdot 60^2)} = 0,213 \text{ A}$$

$$\text{b)} \quad \Delta U_e = P \cdot \Delta t = (\Delta V \cdot A) \cdot \Delta t = 1,5 \times 0,213 \times (2 \times 60^2) = 2,30 \text{ kJ}$$

$$\text{c)} \quad P = \Delta V \cdot A = 1,5 \times 0,213 = 0,320 \text{ W}$$

$$\text{d)} \quad q_i = 3 \text{ A} \cdot \text{h}$$

$$q_f = q_i - \bar{I} \cdot t = 3 - 0,213 \times 2 = 2,574 \text{ A} \cdot \text{h}$$

9)

$$E = 732 \text{ kJ}$$

$$\Delta t = 60 \text{ s}$$

$$\Delta V = 220 \text{ V}$$

a)

$$P = \frac{E}{\Delta t} = \frac{732 \text{ kJ}}{60} = 12,2 \text{ kW}$$

$$P = I \cdot \Delta V \Leftrightarrow I = \frac{12,2 \text{ kW}}{220} = 55,45 \text{ A}$$

b)

$$E_{10 \text{ min}} = P \cdot \Delta t_{10 \text{ min}} = 12,2 \text{ kW} \cdot \left(\frac{10}{60} \right) = 2,03 \text{ kWh} = 0,367 \text{ kWh}$$

$$p_{\text{rego}} = 12 \times E_{10 \text{ min}} = 12 \times 0,367 = 4,40 \text{ centimes}$$